Ranking, Calibration and Cost in Competing-Risks Survival Models: A Common-Protocol Benchmark on Mortgage Data

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***Abstract*—Lenders must state how likely a loan is to default by a given date, not merely which loans are riskiest, and a loan can end by default, prepayment or maturity. Deep competing-risks models are usually reported on ranking alone. We benchmark cause-specific Cox, Fine–Gray, a random survival forest and DeepHit on 550,000 Freddie Mac mortgages under one protocol that measures discrimination, calibration and computational cost together, with vintage-ordered splits and a tuning budget fixed in advance at twenty configurations for DeepHit and eight for the forest. DeepHit's advantage depends on the regime: with tuning it beats a cause-specific Cox model by 0.019 AUC on crisis-era loans and loses to it by 0.009 on recent ones. Its calibration penalty does not reverse, and it never beats a random survival forest. A pooled view of 40 tuning configurations appears to show that DeepHit's ranking-loss weight strongly affects both discrimination and calibration; a controlled ablation holding architecture fixed shows the effect is small and reverses sign across cohorts and causes. We also pre-registered and failed to confirm the hypothesis that selecting on discrimination costs calibration.**

I. Introduction

Credit scoring has long been posed as classification at a fixed horizon: will this borrower default within a year? Provisioning rules ask something harder. A lender must estimate losses over the remaining life of a loan, which requires a probability of default at every future date, and loans end in more than one way. A mortgage may default, be repaid early, or run to maturity, and once one of these happens the others cannot. Under such competing risks the quantity needed is the cumulative incidence function of default rather than one minus a survival curve [1]. The distinction is not academic: treating early repayment as missing information biases the estimated default probability upward [2], and a model can order borrowers correctly while attaching the wrong numbers to them.

Deep models that predict a cumulative incidence per cause exist and have been applied to mortgages [3], [4], but they are selected and reported on ranking. Our three literature reviews — of competing risks in credit scoring, of calibration measures, and of evaluation protocol and cost — converged independently on that omission: these models' probabilities are seldom checked, and the protocols used to check anything seldom resemble deployment. This paper asks whether the advantage survives a protocol that measures ranking, calibration and cost together, on 550,000 Freddie Mac mortgages across two economic regimes.

II. Background and Related Work

Two recent reviews bracket this problem without meeting in the middle. Shang et al. [5] survey three decades of survival analysis in credit risk but do not examine how the resulting predictions are checked. Wiegrebe et al. [6] review more than fifty deep learning methods for time-to-event data and observe that most handle only a single risk with censoring; they consider no credit application. Neither asks whether credit models produce faithful probabilities.

On the credit side, the field's benchmark compares eight survival models on ten portfolios and finds that seven of them censor early repayment; on the same loans that choice moves the censoring rate from 20–85% to 94.56–98.16% [7]. Later work adopts competing risks explicitly, with cause-specific Cox models [8], random survival forests [2] and boosting [9], while macroeconomic covariates have long been used in the single-risk setting [10]. Blumenstock et al. [4] benchmarked DeepHit and a random survival forest against cause-specific Cox and Fine–Gray on US mortgages with default and prepayment as competing risks, and reported the highest concordance index for DeepHit in every experiment, evaluated over ten random train/test splits. Our model set and event pair are theirs; what we change is the protocol, and the present paper is therefore a replication under vintage-ordered splits with calibration and cost added rather than a new benchmark design. On the evaluation side, DeepHit's loss combines a likelihood

with a ranking term whose weight is tuned on validation discrimination [3]; Haider et al. [11] define D-calibration for single events, and Sonabend et al. [12] show that a concordance index computed on a predicted distribution depends on an unstated reduction to a single risk. Most recently, Alberge et al. [13] show that existing calibration measures are unsuited to competing risks, introduce two proper measures, and give recalibration methods that improve probabilities while preserving discrimination. These measures have not been applied to credit data, and no study evaluates ranking, calibration and cost for these models together.

Two qualifications matter for how that gap should be read. First, the deployed baseline is not a survival model: reviewing bank disclosures, Bank and Eder [14] find lifetime probabilities of default under IFRS 9 built mainly from transformations of regulatory estimates and rating-based matrix models, with discrete-time multinomial logit the usual device where competing risks are modelled at all. Competing risks have therefore reached production, in a model family absent from the benchmarking literature. Second, calibration is not neglected in practice: supervisory guidance defines it as matching a long-run average default rate, and current IFRS 9 term-structure work still censors competing risks for tractability: Botha and Verster [15] report that 47% of the mortgage spells in their study end in a competing risk, 96% of them early settlements, and treat those spells as right-censored. The gap is thus a mismatch between the quantity the benchmarking literature reports and the quantity deployment requires, not an absence of concern for probabilities. Wycinka finds the same on consumer loans: discrimination separated the specifications far less than their agreement with the empirical cumulative incidence did [16].

III. Research Problem and Question

Our three reviews reached the same literature from different directions. A.S. surveyed competing-risks models in credit scoring and found that early repayment is usually censored rather than modelled as a second outcome. A.E.N. surveyed calibration measures for survival predictions and found that credit studies report discrimination almost to the exclusion of anything else. F.B. surveyed evaluation protocol and cost and found that random train/test splits dominate and that fitting cost is seldom reported. These are three symptoms of one condition: deep competing-risks models are chosen and reported on a ranking statistic, under a protocol unlike the way a lender meets a new loan, and their probabilities go unchecked.

The gap follows directly. This research evaluates how much of the reported advantage of deep competing-risks models is a property of the way they are measured rather than of the models themselves. Our research question is: *to what extent does the measured advantage of deep competing-risks models over classical ones change when ranking, calibration and cost are evaluated together on vintage-ordered splits?* It divides into four sub-questions. How does any ranking advantage differ between economic regimes (§V-A)? How much of it survives against a flexible non-deep baseline (§V-B)? What calibration penalty accompanies it, and how much of that penalty can recalibration repair (§V-E)? And how much of a reported advantage is attributable to the evaluation protocol itself (§V-F)?

IV. Protocol

The strategy is experimental in Denscombe's sense: an empirical investigation under controlled conditions producing numerical outcomes, with hypotheses fixed before measurement and controls held over every factor not under test. It is closer to an evaluation design than to a treatment experiment. The data are secondary records of loans rather than of people, so no ethical approval was required; the licence permits academic use but not redistribution, which is why the repository excludes the data and rebuilds the tables from archives the reader downloads.

Four delimitations were fixed before any model was run: US agency mortgages only; covariates observed at origination rather than time-varying macroeconomic series; horizons up to 60 months rather than a full mortgage term; and the lifetime PD term structure alone, one input to expected credit loss, leaving loss given default, exposure at default, discounting and staging out of scope.

A. Data

Our data are 50,000-loan samples per origination year from the Freddie Mac Single-Family Loan-Level Dataset: 300,000 loans originated 2013–2018 and 250,000 originated 2005–2009, with performance observed to March 2026. The post-crisis cohort is 4.5% default and 75.0% prepaid; the crisis cohort, 10.5% and 87.1%. Default is the first month at 90 or more days past due, following [4]; prepayment is zero-balance code 01. Time is loan age in months, so the origin is origination. These data reproduce the contrast that motivates the competing-risks framing: counting prepayment as an event leaves 20.6% of post-crisis loans censored, while censoring it leaves 95.5%. Dirick et al. [7] report 20–85% and 94.56–98.16% on

ten European consumer-credit portfolios; ours sit at the lower edge of the first range and inside the second, on a different product, country and decade. Prepayment is not a nuisance here but the dominant outcome, outnumbering default seventeen to one.

B. Models and the common interface

Every model implements the same interface, taking covariates, times and event labels and returning a predicted cumulative incidence per cause on a fixed horizon grid, so no model can be given a metric the others cannot. We compare the Aalen–Johansen estimator as a null model that cannot discriminate but is calibrated by construction, a cause-specific Cox model, a Fine–Gray subdistribution model, a random survival forest and DeepHit. Two implementation details matter for interpretation. Cox and Fine–Gray were implemented directly, since no mainstream Python library provides Fine–Gray; both maximise a Breslow partial likelihood, with Fine–Gray retaining subjects who experience the competing event in the risk set under the censoring weights of [1]. Our forest is a cause-specific random survival forest, fitting one forest per cause, and is therefore a relative of rather than a replication of the composite-splitting competing-risks forest used by Frydman and Matuszyk [2].

C. Metrics

Discrimination is the inverse-probability-of-censoring-weighted time-dependent AUC at 12 to 60 months, with subjects who experience the competing event treated as controls because they can never experience the cause of interest. Calibration is measured in two ways: the integrated Brier score for the cumulative incidence, and a calibration index comparing mean predicted risk against the Aalen–Johansen estimate within deciles of predicted risk — a binned form of the integrated calibration index [17], whose competing-risks version smooths the curve rather than binning it [18]. Cost is measured fit and prediction time. Censoring weights come from a Kaplan–Meier estimate fitted on the training set only; because the time origin is origination and every vintage has at least eight years of follow-up, $G(60) = 0.9986$ in both cohorts, so those weights are near one and it is the competing event, not censoring, that shapes the results.

D. Splits, tuning budget and hypotheses

Models are trained on earlier origination years and tested on later ones, which is how a lender uses them, in contrast to the random splits that dominate the literature [4]. The post-crisis cohort trains on 2013–2015 and tests on 2016–2018; the crisis cohort trains on 2005–2006 and tests on 2007–2009. Hyperparameters were searched with a budget fixed in advance: twenty random configurations for DeepHit and eight for the forest, scored on an out-of-time validation vintage held inside the training years, so test vintages were never touched during tuning. Cox and Fine–Gray have no comparable hyperparameters and are reported untuned. Each configuration was scored on validation AUC and on validation integrated Brier score, and the winner under each criterion was refitted and evaluated separately.

TABLE I
DISCRIMINATION (AUC AT 60 MONTHS) AND CALIBRATION (DECILE-BINNED ICI AT 60 MONTHS) FOR DEFAULT, ON THE HELD-OUT TEST VINTAGES OF EACH COHORT.

	Crisis (2007–09)		Post-crisis (2016–18)	
Model	AUC	ICI	AUC	ICI
Aalen–Johansen	0.5000	0.0077	–	–
Cox	0.8346	0.0169	0.7372	0.0278
Fine–Gray	0.8327	0.0159	–	–
RSF	0.8516	0.0286	0.7300	0.0287
RSF (tuned)	0.8538	0.0267	0.7275	0.0289
DeepHit α=0.2	0.8301	0.0281	0.7239	0.0242
DeepHit α=1	0.8405	0.0240	0.7246	0.0318
DeepHit (tuned)	0.8538	0.0209	0.7280	0.0318

For inference we fixed, before examining any test result, a two-sided null hypothesis that the models do not differ, a significance level of 0.05, and a Bonferroni correction across the three metric families, giving a threshold of 0.0167. A family is the three metrics of one pre-specified comparison in one cohort; thirty-six tests are computed in all, so the correction holds within a family, not across the table. Differences are assessed by a paired bootstrap rather than a parametric test, whose assumptions fail here: observations are paired, since the quantities compared come from the same test loans, and the sampling distributions of AUC and the calibration index are neither normal nor known in closed form. With 500 resamples it cannot resolve p below $1/500$; 0.002 is an upper bound. Because the bootstrap holds the fitted model fixed and understates total uncertainty, DeepHit was refitted with five seeds per cohort; its seed-to-seed standard deviation was 0.0014 and 0.0010 in AUC, and we treat differences below twice that as indistinguishable regardless of p.

V. RESULTS

A. The advantage reverses with regime

DeepHit's standing depends on which economic regime it is tested in (Table I). On the crisis cohort

the tuned model reaches an AUC of 0.8538 against 0.8346 for the cause-specific Cox model, a difference of $+0.0192$ that is significant under the paired bootstrap. That margin should be read with care: refitting on five bootstrap resamples of the training vintages gives the Cox model a standard deviation of 0.0159 in this cohort and Fine–Gray 0.0206, against 0.0035 for DeepHit and 0.0009 for the forest, so the gap is little more than one standard deviation of the baseline's own training variability: here the classical models are the unstable ones. On the post-crisis cohort the ordering reverses, 0.7280 against 0.7372, a difference of -0.0091 that is also significant (Table III) and, with every training standard deviation below 0.0022 there, outside the noise. Calibration does not reverse: DeepHit's index is 0.0040 worse than Cox's in both cohorts (Fig. 1, left). The two carry that with very different force, about eighteen times the models' combined training variability post-crisis against 1.7 times on crisis data, so the sign replicates and the size does not. Every curve in the figure lies above the diagonal: calibration here is limited less by model class than by the vintage shift.

B. Nothing is gained over a random survival forest

Against a tuned random survival forest, DeepHit offers nothing. The two are statistically indistinguishable on discrimination in both cohorts, with $p = 0.968$ on crisis data and $p = 0.788$ post-crisis, and both differences fall inside DeepHit's own seed variation. These are among the few comparisons in this study where we fail to reject: whatever the deep model captures, a forest captures as well. On prepayment in the crisis cohort the forest is distinctly better, reaching an AUC of 0.685 against 0.620 and a calibration index of 0.154 against 0.236 — the competing event the credit literature most often censors (Fig. 1, right).

C. The loss weight is not a reliable knob

DeepHit's loss is a convex combination α·likelihood $+ (1 - \alpha)$·ranking, and the original model selects this weight on validation discrimination [3]. Pooling our forty tuning configurations by α suggests a strong monotone effect, mean validation AUC rising from 0.658 at $\alpha = 0.1$ to 0.776 at $\alpha = 1.0$, but that is a marginal over a search in which α varied alongside depth, dropout, learning rate, batch size and bin count, and it does not survive control.

Holding everything else fixed and varying only α, over five seeds and both cohorts, the effect is small and inconsistent in sign (Table II). On post-crisis default it is invisible in discrimination, $+0.0005$ against a seed standard deviation of 0.0018. On crisis prepayment removing the ranking term costs 0.067 of AUC and worsens calibration; on post-crisis prepayment it *improves* both. Three of the four cause-by-cohort cells show a discrimination effect larger than seed noise, and they disagree on direction.

TABLE II

CONTROLLED ABLATION OF DEEPHIT'S LOSS WEIGHT α: ARCHITECTURE, LEARNING RATE, BATCH SIZE, DROPOUT AND BIN COUNT HELD FIXED, FIVE SEEDS PER SETTING, EVALUATED AT 60 MONTHS ON THE HELD-OUT TEST VINTAGES. $\alpha = 1$ REMOVES THE RANKING TERM ENTIRELY. THE EFFECT REVERSES SIGN ACROSS COHORTS AND CAUSES, AND IN ONE CELL IS SMALLER THAN THE SEED STANDARD DEVIATION. Δ IS THE CHANGE FROM α=0.1 TO α=1.

Cause	Cohort	ΔAUC	seed sd	ΔICI
def.	crisis	+0.0075	0.0033	-0.0010
def.	post	+0.0005	0.0018	+0.0114
prep.	crisis	-0.0673	0.0103	+0.0345
prep.	post	+0.0386	0.0062	-0.0397

The loss weight is therefore not the dominant hyperparameter the pooled view suggested, and no setting can be recommended from these data. The discrepancy between the two views is itself the result: a marginal taken over a random search is not an ablation.

D. Selecting on discrimination does not cost calibration

We pre-registered the hypothesis that choosing a configuration on validation discrimination would yield a worse calibrated model than choosing on a probability-accuracy criterion, and did not confirm it. Of four model-by-cohort comparisons, three differ by less than twice DeepHit's seed-to-seed variation. The one robust effect is a $+0.0101$ AUC gain in the crisis cohort for the configuration chosen on AUC, which is selection behaving as intended.

E. Recalibration does not help, and why

Since Alberge et al. [13] show that recalibration improves competing-risks probabilities while preserving discrimination, we applied a marginal per-horizon rescaling in that spirit, estimating it two ways (Table IV). Fitted on the training vintages, which is all a lender has when scoring a new loan, it changes almost nothing: in ten of sixteen cells it leaves the calibration index worse than the raw predictions. Fitted instead on the test vintages — an oracle no lender could build — it reduces the index by a median factor of 4.2, reaching 14.7 for Fine–Gray on post-crisis default.

The contrast identifies the penalty. It is not a defective calibration map, or the oracle correction would not work;

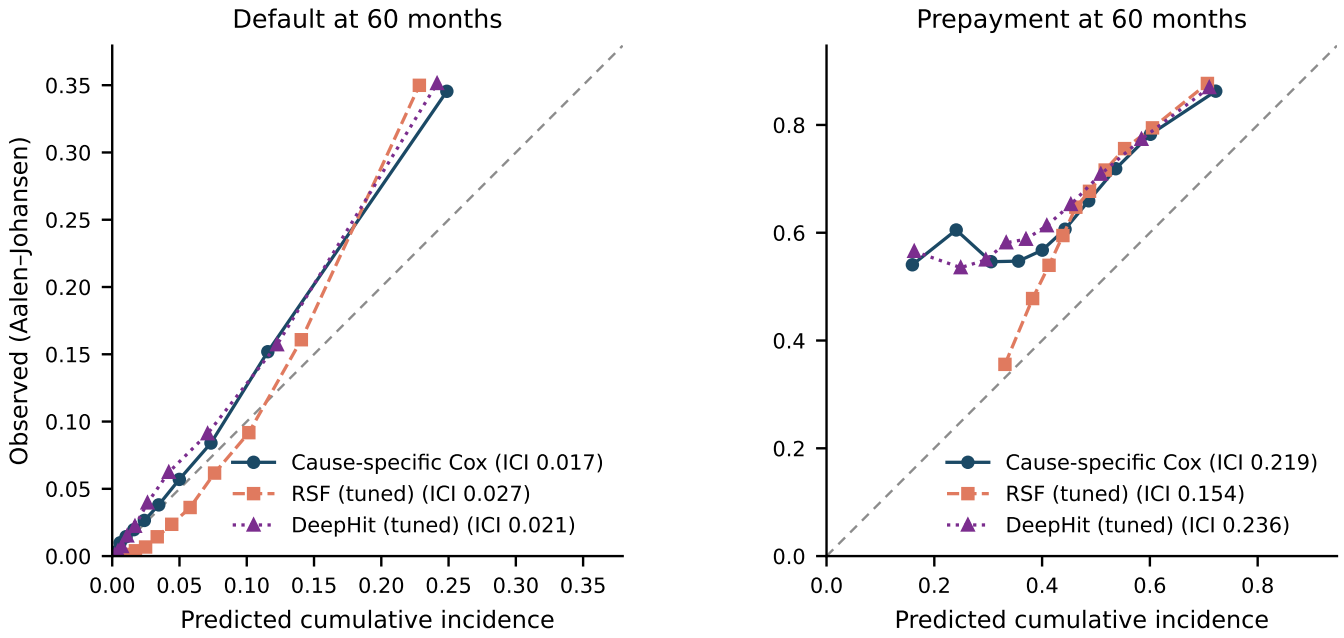


Fig. 1. Calibration at 60 months on the crisis cohort (train 2005–06, test 2007–09). Each point is a decile of predicted risk; the vertical axis is the Aalen–Johansen estimate within that decile, so competing events and censoring are both accounted for. A model on the diagonal is perfectly calibrated; every curve lies above it, so all three under-predict out of time. The deciles separate by a few percentage points for default and by twenty or more for prepayment, the competing event the credit literature most often censors.

TABLE III
PAIRED BOOTSTRAP ON THE TUNED MODELS (B=500), DEFAULT, 60 MONTHS. T1 DEEPHIT VS COX; T2 DEEPHIT VS RSF; T3 DEEPHIT SELECTED ON AUC VS ON IBS. * SIGNIFICANT AT THE BONFERRONI LEVEL $0.05/3$; † DIFFERENCE SMALLER THAN TWICE DEEPHIT'S SEED-TO-SEED SD.

Test	Cohort	Metric	Diff	95% CI	p	
T1	post	AUC	-0.0091	[-0.0116, -0.0066]	<0.002	*
T1	post	ICI	+0.0040	[+0.0040, +0.0041]	<0.002	*
T2	post	AUC	+0.0005	[-0.0024, +0.0038]	0.788	†
T2	post	ICI	+0.0030	[+0.0029, +0.0030]	<0.002	*
T3	post	AUC	+0.0005	[-0.0017, +0.0026]	0.604	†
T3	post	ICI	+0.0015	[+0.0014, +0.0015]	<0.002	*
T1	crisis	AUC	+0.0192	[+0.0167, +0.0214]	<0.002	*
T1	crisis	ICI	+0.0040	[+0.0038, +0.0043]	<0.002	*
T2	crisis	AUC	-0.0000	[-0.0019, +0.0020]	0.968	†
T2	crisis	ICI	-0.0057	[-0.0069, -0.0041]	<0.002	*
T3	crisis	AUC	+0.0101	[+0.0091, +0.0112]	<0.002	*
T3	crisis	ICI	+0.0019	[+0.0017, +0.0022]	<0.002	†

TABLE IV
POST-HOC RECALIBRATION, ICI AT 60 MONTHS. "TRAIN" ESTIMATES THE RESCALING ON THE TRAINING VINTAGES, WHICH IS ALL A LENDER HAS AT SCORING TIME; "ORACLE" ESTIMATES IT ON THE TEST VINTAGES AND IS UNATTAINABLE IN PRACTICE. COX AND DEEPHIT ARE SHOWN; THE FOREST AND FINE–GRAY BEHAVE THE SAME WAY. AUC IS IDENTICAL TO FOUR DECIMALS FOR RAW AND RECALIBRATED PREDICTIONS IN ALL SIXTEEN CELLS, CONFIRMING THE RESCALING IS MONOTONE.

Cohort	Cause	Model	Raw	Recal. train	Recal. oracle
crisis	def.	Cox	0.0169	0.0182	0.0061
crisis	def.	DeepHit	0.0209	0.0164	0.0053
crisis	prep.	Cox	0.2186	0.2204	0.1168
crisis	prep.	DeepHit	0.2361	0.2222	0.1063
post	def.	Cox	0.0278	0.0279	0.0025
post	def.	DeepHit	0.0318	0.0297	0.0082
post	prep.	Cox	0.1640	0.1693	0.0253
post	prep.	DeepHit	0.1850	0.1685	0.0439

it is the vintage shift, since a rescaling estimated on 2005–06 cannot know that 2007–09 would turn out worse. Recalibration as demonstrated in the literature is an in-distribution procedure, and extending it across vintages is an open problem rather than a step we omitted.

F. What the evaluation protocol is worth

The benchmarking literature evaluates on random splits; we evaluate on vintage-ordered ones. Holding models and tuned settings fixed and changing only how the test set is drawn isolates what that choice is worth (Table V). Under random splits DeepHit leads both baselines in all eight cause-by-cohort cells, by $+0.0022$ to $+0.0178$ AUC, with no seed standard deviation above 0.0019. Under vintage ordering three of those leads

TABLE V
WHAT THE EVALUATION PROTOCOL IS WORTH. DEEPHIT'S AUC ADVANTAGE AT 60 MONTHS OVER THE CAUSE-SPECIFIC COX MODEL AND OVER THE TUNED FOREST, MEASURED UNDER RANDOM SPLITS (MEAN OF FIVE SEEDS, MATCHED TO THE VINTAGE SPLIT'S TRAIN AND TEST SIZES; EVERY SEED STANDARD DEVIATION IS AT OR BELOW 0.0019) AND UNDER VINTAGE-ORDERED SPLITS. MODELS AND TUNED SETTINGS ARE IDENTICAL THROUGHOUT; ONLY THE WAY THE TEST SET IS CHOSEN CHANGES. UNDER RANDOM SPLITS DEEPHIT LEADS IN ALL EIGHT CELLS; UNDER VINTAGE ORDERING THREE OF THOSE LEADS REVERSE.

		vs Cox		vs RSF	
Cause	Cohort	random	vintage	random	vintage
def.	crisis	+0.0025	+0.0192	+0.0032	-0.0000
def.	post	+0.0022	-0.0091	+0.0035	+0.0005
prep.	crisis	+0.0178	+0.0012	+0.0104	-0.0651
prep.	post	+0.0139	+0.0012	+0.0108	+0.0048

reverse: DeepHit falls behind Cox on post-crisis default, draws level with the forest on crisis default, and loses to it on crisis prepayment by 0.065.

The mechanism is not only leakage. A random split both places temporal neighbours of each test loan in the training set and removes the shift between origination vintages, and a lender faces neither advantage; the two cannot be separated here. Absolute scores do not move in one direction: on post-crisis default every model gains $+0.021$ to $+0.033$ AUC, while on crisis default the flexible models lose, 2007–09 carrying more default signal than a random draw. What moves consistently is the comparison between models, and it moves in DeepHit's favour. The tuned configurations were selected on out-of-time validation, so the effect is a lower bound.

G. Cost

The models span three orders of magnitude in training cost. On 150,000 loans, Fine–Gray fits in 0.03 seconds and the cause-specific Cox model in 0.42, against 36 seconds for the forest and 59 for DeepHit, and the DeepHit search added 14.4 minutes of fitting across the two cohorts and the forest search 4.1, while the classical models have no search to run. Since the deep model is indistinguishable from the forest and loses to Cox in one of the two regimes, that factor of roughly two thousand buys no reliable improvement. We report these figures because the benchmarking literature generally omits them [6].

VI. DISCUSSION, LIMITATIONS AND CONCLUSION

Taken together these results qualify, rather than contradict, the finding that deep competing-risks models perform well on mortgage data [4]. DeepHit does win on crisis-era loans once tuned, and an untuned deep baseline is not evidence. But the win is confined to one regime, disappears against a random survival forest, costs three orders of magnitude more compute, and carries a calibration penalty throughout. For a lender needing a probability rather than an ordering, that is difficult to justify.

Against the quality criteria for quantitative work, reliability is measured rather than asserted: every comparison carries a variability estimate from seeds, from bootstrap resamples of the test set, and from refitting on resampled training vintages. Replicability is the public pipeline. Internal validity rests on varying one factor at a time, which the α ablation and the protocol comparison each enforce. Construct validity is the paper's subject rather than an assumption: we ask whether discrimination measures what it is used to decide. Generalisation is weakest, and bounded by the first delimitation above.

Within those bounds, several limitations remain. Our recalibration is a marginal per-horizon rescaling rather than the per-cause procedures of Alberge et al. [13]. Training variability is estimated from five bootstrap resamples, which conflates sampling with optimisation noise. The forest is cause-specific rather than Ishwaran's competing-risks formulation. And excluding the time-varying macroeconomic variables of Bellotti and Crook [10] leaves the vintage-shift result without a mechanism.

Under one protocol measuring ranking, calibration and cost together on vintage-ordered splits, deep competing-risks models offer no reliable advantage over classical or tree-based baselines on mortgage data, are worse calibrated in both regimes, and depend on the regime for what ranking advantage they show. What generalises past this benchmark is methodological: a hyperparameter's apparent effect in a pooled search did not survive controlled variation.

AUTHOR CONTRIBUTIONS

The three reviews were written independently: A.S. on competing risks in credit scoring, A.E.N. on calibration, F.B. on evaluation protocol and cost. A.S. built the pipeline, models and metrics and ran the experiments; A.E.N. specified the calibration measures and recalibration study; F.B. the split protocol, tuning budget and cost measurement. The gap, hypotheses and final analysis were agreed jointly; code at github.com/ar-sadraei/rtds-project.

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